

Lale Özkahya

ozkahya@cs.hacettepe.edu.tr

<http://ozkahya.github.io>

POSITIONS	Department of Artificial Intelligence and Data Engineering at Hacettepe University	
	Associate Professor	2026–
	Dept. of Computer Engineering at Hacettepe University	
	Associate Professor	2014-2026
	Dept. of Mathematics at Hacettepe University	
	Expert	2012-2014
	Dept. of Mathematics at Işık University	
	Postdoctoral researcher, Supervisor: Türker Bıyıkoglu	Jan.-May 2012
	Dept. of Mathematics at Iowa State University	
	Postdoctoral researcher, Supervisor: Maria Axenovich	Aug. 2011- May 2012
	Institute of Pure and Applied Mathematics (IPAM)	
	<i>Combinatorics: Methods and App. in Math. and Comp. Sci.</i> Program	
	Long-term sponsored fellow	Sep.-Dec. 2009
EDUCATION	Dept. of Mathematics at University of Illinois	
	Graduate researcher	2004 - 2009
	Dept. of Aerospace engineering at University of Illinois	
	Graduate researcher	2001 - 2003
EDUCATION	Ph.D. Mathematics , University of Illinois, Advisor: Zoltán Füredi	2010
	M.S. Aerospace engineering , (Computational Science and Engineering Program), University of Illinois, Advisor: Philippe Geubelle	2003
	B.S. Mechanical engineering , Boğaziçi (Bosphorus) University	1999
RESEARCH INTERESTS	Network analysis, graph mining, extremal graph theory	
AWARDS& HONORS	Science Academy Young Scientist Award (BAGEP, Mathematics)	2018
	Invited Participant at Geometric and Enumerative Combinatorics at IMA	2014
	Invited Participant at Hypergraph Turan problem Workshop at AIM	2011
JOURNALS	[1] L. Özkahya, P. Sarıyerli: "Star Coloring of Hypergraphs", submitted.	
	[2] D. Keskinçiliç, L. Özkahya: "Coloring Grids Avoiding Bicolored Paths", <i>Discrete Mathematics & Theoretical Computer Science</i> 28 (3) (2026).	
	[3] A. Kırtısoğlu, L. Özkahya: "Coloring of Graphs Avoiding Bicolored Paths of a Fixed Length", <i>Graphs and Combinatorics</i> 40 (1) (2024), Article 11.	
	[4] L. Özkahya, Y. Person: "Minimum Rainbow H -decompositions of graphs", <i>European Journal of Combinatorics</i> 71 (2018), 111-124.	
	[5] Z. Füredi, L. Özkahya: "On 3-uniform hypergraphs without a cycle of a given length", <i>Discrete Applied Mathematics</i> 216 (3) (2017), 582–588.	
	[6] L. Özkahya, B. Stanton: "On a covering problem in the hypercube", <i>Graphs and Combinatorics</i> 31 (1) (2015), 235–242.	

- [7] L. Özkahya, M. Young: “Anti-Ramsey number of matchings in hypergraphs”, *Discrete Mathematics* **313**(20) (2013), 2359–2364 .
- [8] M. Axenovich, L. Özkahya: “On homometric sets in graphs”, *Australas. J. Combin.* **55** (2013), 175–187.
- [9] Z. L. Nagy, L. Özkahya, B. Patkós, M. Vizer: “On the ratio of maximum and minimum degree in maximal intersecting families”, *Discrete Mathematics* **313** (2013), 207–211.
- [10] L. Özkahya, Y. Person: “Minimum H -decompositions of graphs: edge-critical case”, *Journal of Combinatorial Theory, Series B*, **102** (2012), 715–725.
- [11] Z. Füredi, L. Özkahya: “Unavoidable subhypergraphs: α -clusters”, *Journal of Combinatorial Theory, Series A* **118** (2011), 2246–2256.
- [12] Z. Füredi, L. Özkahya: “On even-cycle-free subgraphs of the hypercube”, *Journal of Combinatorial Theory, Series A* **118** (2011), 1816–1819.
- [13] J. Choi, L. Özkahya, D. West: “Degree-splittability of caterpillars and regular multigraphs”, *Congressus Numerantium* **202** (2010), 137–148.
- [14] A. Kostochka, L. Özkahya and D. Woodall: “A Brooks-type bound for squares of K_4 -minor-free graphs”, *Discrete Mathematics* **309** (2009), 6572–6584.
- [15] Z. Füredi, L. Özkahya: “On 14-cycle-free subgraphs of the hypercube”, *Combinatorics, Probability and Computing* **18** (2009), 725–729.

CONFERENCE PUBLICATIONS

- [16] Ali Baran Taşdemir, Emir Kaan Kırmacı , Tuna Karacan, Lale Özkahya. Learning Heuristics for the Maximum Clique Enumeration Problem Using Low Dimensional Representations. The 11th International Conference on Computational Data and Social Networks (**CSoNet2022**), 9 pages, December 2022.
- [17] Murat Çelik, Ali Baran Taşdemir, Lale Özkahya. Low-Rank Representations Towards Classification Problem of Complex Networks. Signal Processing and Communications Applications Conference (**SIU2022**), 4 pages, May 2022.
- [18] Ali Baran Taşdemir, Lale Özkahya. Learning Heuristics for the Maximum Clique Enumeration Problem Using Higher-Order Graph Structures. Signal Processing and Communications Applications Conference (**SIU2021**), 4 pages, June 2021.
- [19] Ali Baran Taşdemir, Barkın Atasay, Lale Özkahya. Classification of Complex Networks Using Structural Analysis of Random Graph Models. Signal Processing and Communications Applications Conference (**SIU2020**), 4 pages, October 2020.
- [20] Taha Sevim, Muhammet Selçuk Güvel, Lale Özkahya. A Fast Counting Method for 6-Motifs with Low Connectivity Mathematical Aspects of Computer and Information Sciences (**MACIS 2019**), 8 pages, November 2019.

TEACHING EXPERIENCE

Undergraduate Teaching

- AIN312 Formal Languages and Automata Theory 2026-
- BBM205 Discrete Mathematics 2015-
- BBM 301 Programming Languages 2021- 2025
- BBM462 Social and Economic Networks 2022-2023
- BBM402 Theory of Computation 2018-2020, 2024-
- BBM 401 Automata Theory and Formal Languages 2016, 2017
- BBM403 Combinatorics and Graph Theory 2015, 2019

Graduate Teaching

- CMP741 Advanced Analysis of Algorithms 2021-2022
- CMP694 Graph Theory 2015-2020, 2023-

STUDENTS MENTORED

Ph.D. Students

- Derman Keskinçilic (2022-)

M.S. Students

- Ali Baran Taşdemir (2020-2022, PhD student at Hacettepe University)
- Alaitin Kırtışoğlu (2019-2021, PhD student at Illinois Institute of Technology)

PROJECTS

TUBITAK-1002, Maksimum Klik Probleminin Çözümü için Yüksek Dereceli Çizge Motiflerinin Katkısıyla Nokta Sınıflandırılması.

Principal Investigator

Feb-Nov 2021

TUBITAK-1001, 'TARİF : Kendin yap videolarının eşleştirilmesi ve zenginleştirilmesi için çok-kipli yaklaşım'.

Researcher

2019-2022

TUBITAK and European Science Foundation (EUROCORES Programme EUROGIGA), 'Geometric representations and symmetries of graphs, maps and other discrete structures and applications in science' (GReGAS).

Researcher

Jan-May 2012

INVITED TALKS

Erdős Günleri, Bogazici University	2024
Analysis Seminar, Bilkent University	2022
IMBM Discrete Math Days, Boğaziçi University	2019
Institute of Mathematics at Goethe University	2018
Dept. of Mathematics at Koç University	2017
Gwangju Institute of Science and Technology	2015
Workshop on Graph Theory and its Applications III at IMBM.	2012
National Mathematics Symposium at Nigde University	2012
Special Session on Graphs and Hypergraphs at AMS Fall Central Section Meeting	2010
Conference in honor of Laci Babai's 60th birthday Combinatorics, Groups, Algorithms, and Complexity	2010

EXTERNAL SERVICE

Journal Reviewer

Australasian J. of Combin.	2016
Cent. Eur. J. of Math.	2013
Combin. Probab. Comput.	2014
Discrete Applied Math.	2015
Discrete Math.	2011-2018, 2022-2024
Discuss. Math. Graph Theory	2011, 2016

Electronic J. of Combin.	2019, 2022
Graphs and Combin.	2011, 2014, 2024
Hacettepe J. of Math. and Stat.	2013, 2014
IEEE Transactions on Industrial Informatics	2017
J. Graph Theory	2012, 2014, 2020
J. of Combin. Optim.	2016
J. Combin. Theo. Ser. A.	2017
J. Combin. Theo. Ser. B.	2024
Turkish J. of Elec. Eng. Comp. Sci.	2014, 2019, 2021
Turkish J. of Math.	2012, 2013, 2015, 2018

Leading Roles

Co-organizer of <i>Special Session on Graph Theory</i> at AMS Central Section Meeting. Iowa City.	2011
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